

Thermoformable Shell for Repeatable Thermoforming

Donghyeon Ko
donghyeon.ko@kaist.ac.kr
KAIST
Daejeon, South Korea

Myeongseong Kim
pluto6944@kaist.ac.kr
KAIST
Daejeon, South Korea

Woohun Lee
woohun.lee@kaist.ac.kr
KAIST
Daejeon, South Korea

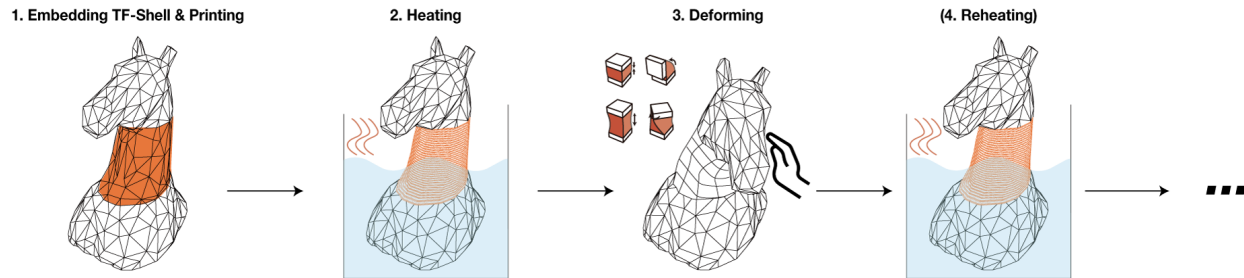


Figure 1: Repeatable thermoforming with *TF-Shell*

ABSTRACT

We propose a thermoformable shell called *TF-Shell* that allows repeatable thermoforming. Due to the low thermal conductivity of typical printing materials like polylactic acid (PLA), thermoforming 3D printed objects is largely limited. Through embedding *TF-Shell*, users can thermoform target parts in diverse ways. Moreover, the deformed structures can be restored by reheating. In this demo, we introduce the *TF-Shell* and demonstrate four thermoforming behaviors with the *TF-Shell* embedded figure. With our approach, we envision bringing the value of hands-on craft to digital fabrication.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

3D printing; thermoforming; metamaterial; self-repair

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1 INTRODUCTION

Although the automation of 3D printing is convenient and precise, users should fabricate 3D objects through virtual modeling. Heat

can be used to soften rigid prints to manually deform them. However, due to the low thermal conductivity of typical FDM printed materials, thermoforming is limited to the deformation thin parts or surface finishing processes such as smoothing. The use of thermoforming in 3D printing can be beneficial because it enables the application of appropriate digital and tangible methods. Users can develop the design while feeling the objects' size or proportions. The fabrication process can be flexible with an additional iteration point, i.e., modification after printing [4, 6, 16].

To facilitate thermoforming, researchers have suggested diverse approaches such as a custom printer to use functional materials [16] or inserting heating elements to heat from the inside [4]. Distinctively, Ko et al. [6] suggested modifying internal structures that pass hot air, so users can deeply and locally heat the printed objects. However, this approach also has several limitations. First, it is difficult for users to carefully control the speed and temperature of the hot air. Additionally, global heating or the heating of large objects is not feasible. Above all, iterative thermoforming is almost impossible as the structures collapse by heating, and it is difficult to restore to their original shape.

We propose a thermoformable shell called *TF-Shell*. Like *TF-Cells* [6], our structure allows the thermoforming of printed objects while addressing the aforementioned issues. Moreover, by applying shape memory, the shell can be restored after reheating, which supports design iterations. As illustrated in Figure 1, if users embed the *TF-Shell* to target parts and print, they can thermoform with hot air or hot water. Upon reheating, they can restore their original shape and keep iterating thermoforming. In this demo, we first introduce the principle of the *TF-Shell*, and demonstrate thermoforming and restoration through the structure.

2 THERMOFORMABLE SHELL

We aimed to use two characteristics of polylactic acid (PLA), which is one of the most widely used filament materials. First, PLA has

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low glass transition temperatures (T_g) (around 60 °C). When heated beyond T_g , the stiffness of the PLA decreases significantly. Thus, it is possible to thermoform 3D printed objects at a relatively safe temperature. Furthermore, the structures do not collapse upon melting, which allows thermoforming multiple times. Additionally, shape memory is one of the characteristics of PLA. If users deform PLA beyond T_g and retain the deformation until it cools down below T_g , a temporary shape can be set. Upon reheating, the temporary shape restores to its original shape [1, 7, 10]. Thus far, many researchers have applied this characteristic to investigate 4D printing such as folding [2, 12, 14, 15], shrinking [13], and expanding [5] into pre-programmed forms. Referring to their techniques, we use the shape memory nature to make thermoforming repeatable, i.e., restoring to original shapes from the deformed shapes for design iterations.

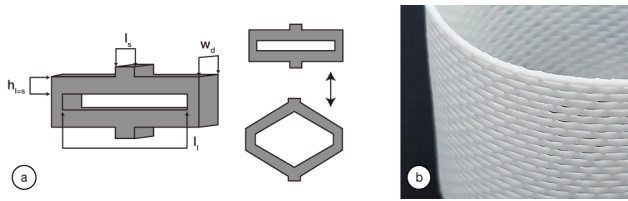


Figure 2: *TF-Shell*: (a) unit cell, (b) printed version

We designed the *TF-Shell* considering the following criteria: high heat transference, flexibility at the target temperature, applicability to most 3D geometries, and printability. The designed *TF-Shell* is a perforated shell structure consisting of repeating structural units. As illustrated in Figure 2.(a), each unit comprises a long bridge (length: l_b) with a short column placed at the center of the bridge. As 3D models are sliced into shell and infill structures in typical slicing tools, the shell structures are compatible with most 3D geometries without lowering printability. Thin and perforated structures have high heat transfer from air or water. When deformed, the long bridge is pulled by the short column and thus bent into a diamond-like structure which affords large strain (Figure 2. (a) right). Therefore, the shell-embedded parts become thermoformable. When reheating to T_g , the deformed shapes of the printed objects can return to their original shape with no structural constraints, e.g., less-heated massive parts like the infill structure.

3 THERMOFORMING THROUGH THE *TF-SHELL*

With the *TF-Shell*, diverse thermoforming behaviors, such as bending, twisting, stretching, and pressing, are applicable. We first demonstrate the four deformation behaviors and their restorations with the cuboid specimens. We then showcase thermoforming of a more complex geometry, a horse figure.

In practical scenarios, the *TF-Shell* is embedded into volumetric body surfaces and thermoformed. We printed 40 mm × 40 mm × 60 mm cuboids comprising the *TF-Shell*, where the top and bottom were solid. We connected a 10 N weight for bending, pressing, and stretching. For twisting, a NEMA 17 step motor (0.45 Nm maximum torque) was used. We immersed specimens at 65 °C for 1 min, then applied force for 30 s. After each deformation, we re-immersed deformed specimens in the same water tank until they were fully



Figure 3: Thermoforming behaviors and restorations: bending, twisting, stretching, and pressing

restored. The room temperature was maintained at 24 °C by an air conditioner.

As illustrated in Figure 3, we could bend by more than 90 °, twist 287.3 ° (SD: 4.7 °), stretch 0.4 strain (SD: 0.04), and press -0.75 strain (SD: 0.01). Upon reheating, every cuboid restored more than 99.7 % within 15 s (twisted cuboids: 99.7 % within 10 s, pressed cuboids: 99.7 % within 14 s, stretched cuboids: 100 % within 3 s, and bent cuboids: 100 % within 3 s)

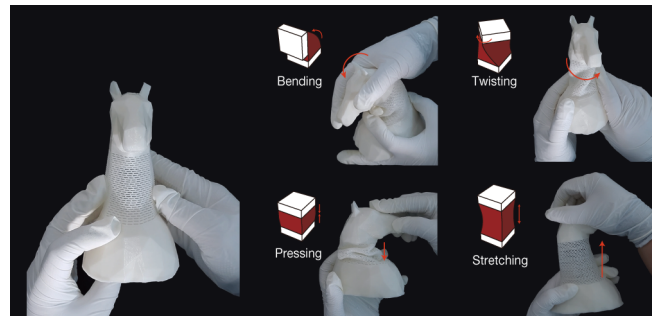


Figure 4: Thermoforming *TF-Shell* embedded horse

Next, we embedded the *TF-Shell* into the neck of the horse figure, downloaded from thingiverse [3]. Compared with cuboids, their shape is more complex, as each layer consists of a polygon with an arbitrary number of sides. The size of the figure was approximately 85 mm × 120 mm × 150 mm.

As illustrated in Figure 4, we could easily heat and deform the shell-embedded horses for four deformations. Additionally, the thermoformed objects can be restored after reheating.

4 CONCLUSION AND FUTURE WORK

To facilitate thermoforming in 3D printing, we presented *TF-Shell*. *TF-Shell* can be embedded into volumetric geometries, thus affording diverse thermoforming processes such as bending, twisting, stretching, and pressing. Moreover, upon reheating, the thermoformed objects can return to their original shape, which supports design iteration. We demonstrated thermoforming and restoration using the *TF-Shell* with cuboids and a horse figure. In the future, we will elaborate on technical evaluations, develop a design tool, and present applicable contexts such as mechanical fitting, aesthetic tuning, and design iteration. Finally, we believe our approach could provide a way to integrate hands-on craft into digital fabrication while adding the values of the crafts such as creative engagement, uniqueness of the final product, and enjoyment [8, 9, 11].

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